

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

Two moles a monoatomic gas is mixed with six moles of a diatomic gas. The molar specific heat of the mixture at constant volume is :

(1) $\frac{9}{4}R$

(2) $\frac{7}{4}R$

(3) $\frac{3}{2}R$

(4) $\frac{5}{2}R$

Q2 - 2024 (01 Feb Shift 2)

If the root mean square velocity of hydrogen molecule at a given temperature and pressure is 2 km/s, the root mean square velocity of oxygen at the same condition in km/s is :

(1) 2.0

(2) 0.5

(3) 1.5

(4) 1.0

Q3 - 2024 (27 Jan Shift 1)

The average kinetic energy of a monatomic molecule is 0.414eV at temperature :

(Use $K_B = 1.38 \times 10^{-23} \text{ J/mol} - \text{K}$)

(1) 3000 K

(2) 3200 K

(3) 1600 K

(4) 1500 K

Q4 - 2024 (27 Jan Shift 2)

The total kinetic energy of 1 mole of oxygen at 27°C is :

[Use universal gas constant (R) = 8.31 J/moleK]

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

(1) 6845.5 J

(2) 5942.0 J

(3) 6232.5 J

(4) 5670.5 J

Q5 - 2024 (29 Jan Shift 1)

Two vessels A and B are of the same size and are at same temperature. A contains 1 g of hydrogen and B contains 1 g of oxygen. P_A and P_B are the pressures of the gases in A and B respectively, then $\frac{P_A}{P_B}$ is :

(1) 16

(2) 8

(3) 4

(4) 32

Q6 - 2024 (29 Jan Shift 2)

The temperature of a gas having 2.0×10^{25} molecules per cubic meter at 1.38 atm (Given, $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$) is :

(1) 500 K

(2) 200 K

(3) 100 K

(4) 300 K

Q7 - 2024 (29 Jan Shift 2)

N moles of a polyatomic gas ($f = 6$) must be mixed with two moles of a monoatomic gas so that the mixture behaves as a diatomic gas. The value of N is :

(1) 6

(2) 3

(3) 4

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

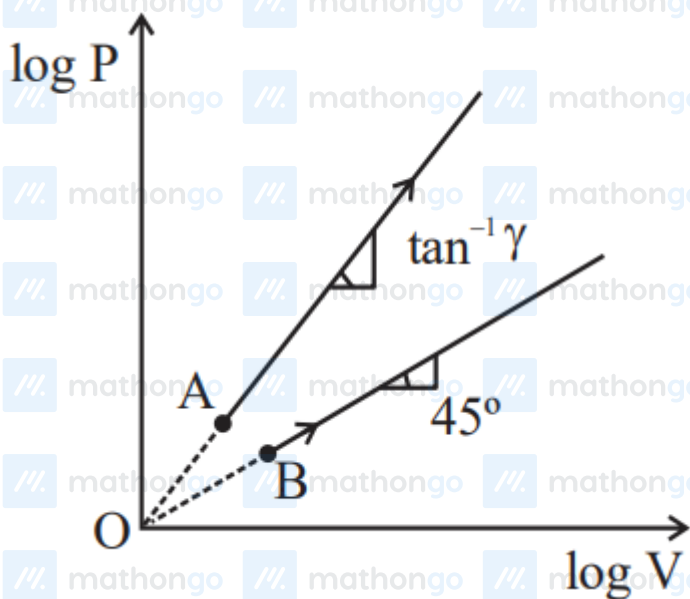
Questions with Answer Keys

MathonGo

(4) 2

Q8 - 2024 (30 Jan Shift 1)

Two thermodynamical process are shown in the figure. The molar heat capacity for process A and B are C_A and C_B . The molar heat capacity at constant pressure and constant volume are represented by C_P and C_V , respectively. Choose the correct statement.



[We changed options. In official NTA paper no option was correct.]

(1) $C_B = \infty, C_A = 0$

(2) $C_A = 0$ and $C_B = \infty$

(3) $C_P > C_B > C_A > C_V$

(4) $C_A > C_P > C_V$

Q9 - 2024 (30 Jan Shift 1)

At which temperature the r.m.s. velocity of a hydrogen molecule equal to that of an oxygen molecule at 47°C ?

(1) 80 K

(2) -73 K

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

(3) 4 K

(4) 20 K

Q10 - 2024 (30 Jan Shift 2)

If three moles of monoatomic gas ($\gamma = \frac{5}{3}$) is mixed with two moles of a diatomic gas ($\gamma = \frac{7}{5}$), the value of adiabatic exponent γ for the mixture is:

(1) 1.75

(2) 1.40

(3) 1.52

(4) 1.35

Q11 - 2024 (31 Jan Shift 1)

The parameter that remains the same for molecules of all gases at a given temperature is :

(1) kinetic energy

(2) momentum

(3) mass

(4) speed

Q12 - 2024 (31 Jan Shift 2)

A gas mixture consists of 8 moles of argon and 6 moles of oxygen at temperature T. Neglecting all vibrational modes, the total internal energy of the system is

(1) 29RT

(2) 20RT

(3) 27RT

(4) 21RT

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Answer Key

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Q1 (1) mathongo /// mathon**Q2** (2) /// mathon**Q3** (2) mathongo /// mathon**Q4** (3) /// mathongo

Q5 (1) mathongo /// mathon**Q6** (1) /// mathon**Q7** (3) mathongo /// mathon**Q8** (3) /// mathongo

Q9 (4) mathongo /// mathon**Q10** (3) /// mathon**Q11** (1) mathongo /// mathon**Q12** (3) /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

Q1

$$C_V = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2}$$

$$= \frac{2 \times \frac{3}{2}R + 6 \times \frac{5}{2}R}{2 + 6}$$

$$= \frac{9}{4}R$$

Q2

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{M_2}{M_1}} \Rightarrow \frac{2}{V_2} = \sqrt{\frac{32}{2}}$$

$$V_2 = 0.5 \text{ km/s}$$

Q3

For monoatomic molecule degree of freedom = 3.

$$\therefore K_{\text{avg}} = \frac{3}{2} K_B T$$

$$T = \frac{0.414 \times 1.6 \times 10^{-19} \times 2}{3 \times 1.38 \times 10^{-23}}$$

$$= 3200 \text{ K}$$

Q4

$$\text{Kinetic energy} = \frac{f}{2} nRT$$

$$= \frac{5}{2} \times 1 \times 8.31 \times 300 \text{ J}$$

$$= 6232.5 \text{ J}$$

Q5

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$\frac{P_A V_A}{P_B V_B} = \frac{n_A R T_A}{n_B R T_B}$$

Given $V_A = V_B$ And $T_A = T_B$

$$\frac{P_A}{P_B} = \frac{n_A}{n_B}$$

$$\frac{P_A}{P_B} = \frac{1/2}{1/32} = 16$$

Q6

$$PV = nRT$$

$$PV = \frac{N}{N_A} RT$$

N = Total no. of molecules

$$P = \frac{N}{V} kT$$

$$1.38 \times 1.01 \times 10^5 = 2 \times 10^{25} \times 1.38 \times 10^{-23} \times T$$

$$1.01 \times 10^5 = 2 \times 10^2 \times T$$

$$T = \frac{1.01 \times 10^3}{2} \approx 500 \text{ K}$$

Q7

$$f_{eq} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$$

For diatomic gas $f_{eq} = 5$

$$5 = \frac{(N)(6) + (2)(3)}{N + 2}$$

$$5N + 10 = 6N + 6$$

$$N = 4$$

Q8

For process A

$$\log P = \gamma \log V \Rightarrow P = V^\gamma, (\gamma > 1)$$

$$PV^{-\gamma} = \text{Constant}$$

$$C_A = C_V + \frac{R}{1+\gamma} \dots (i)$$

Likewise for process B $\rightarrow PV^{-1} = \text{Cons tan } t$

$$C_B = C_v + \frac{R}{1+1}$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$C_B = C_v + \frac{R}{2} \dots (ii)$$

$$C_P = C_v + R \dots (iii)$$

By (i), (ii) & (iii)

$$C_P > C_B > C_A > C_v \text{ [No answer matching]}$$

Q9

$$\sqrt{\frac{3RT}{2}} = \sqrt{\frac{3R(320)}{32}}$$

$$T = \frac{320}{16} = 20 \text{ K}$$

Q10

$$f_1 = 3, f_2 = 5$$

$$n_1 = 3, n_2 = 2$$

$$f_{\text{mixture}} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} = \frac{9 + 10}{5} = \frac{19}{5}$$

$$\gamma_{\text{mixture}} = 1 + \frac{2 \times 5}{19} = \frac{29}{19} = 1.52$$

Q11

$$KE = \frac{f}{2} kT$$

Conceptual

Q12

$$U = nC_V T$$

$$\Rightarrow U = n_1 C_{V_1} T + n_2 C_{V_2} T$$

$$\Rightarrow 8 \times \frac{3R}{2} \times T + 6 \times \frac{5R}{2} \times T$$

$$= 27RT$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)